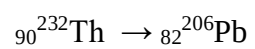


ASSIGNMENT (Chemical Kinetics)

- The rate of decomposition of $\text{NH}_3(\text{g})$ at 10 atm on platinum surface is zero. What is rate of formation (in M min^{-1}) of $\text{H}_2(\text{g})$, if rate constant of reaction $2\text{NH}_3(\text{g}) \rightarrow \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$ is 2.0 M min^{-1} ?
- $5\text{A} \rightarrow \text{Product}$
In above reaction, half-life period is directly proportional to initial concentration of reactant. The initial rate of reaction is $400 \text{ mol lit}^{-1} \text{ min}^{-1}$.
Calculate the half-life period (in sec) when initial concentration of reactant is 200 mol lit^{-1} .
- The rate of decomposition of $\text{NH}_3(\text{g})$ at 10 atm on platinum surface is zero order. What is rate of formation (in M min^{-1}) of $\text{H}_2(\text{g})$ if rate constant of reaction $2\text{NH}_3(\text{g}) \rightarrow \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$ is 2.0 M min^{-1} .
- Carbon monoxide reacts with O_2 to form CO_2 : $2\text{CO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g})$ Information on this reaction is given in the table below:
- Half-life for the zero order reaction $\text{A}(\text{g}) \rightarrow \text{B}(\text{g}) + \text{C}(\text{g})$ and half-life for the first order reaction $\text{X}(\text{g}) \rightarrow \text{Y}(\text{g}) + \text{Z}(\text{g})$ are equal. If completion time for the zero order reaction is 13.86 min, then calculate the rate constant (in hr^{-1}) for reaction $\text{X}(\text{g}) \rightarrow \text{Y}(\text{g}) + \text{Z}(\text{g})$.
- For any acid catalyzed reaction, $\text{A} \xrightarrow{\text{H}^+} \text{B}$ Half-life period is independent of concentration of A at given pH. At same concentration of A half-life time is 10 min at $\text{pH} = 2$ and half life time is 100 min at $\text{pH} = 3$. If the rate law expression of reaction is $r = k[\text{A}]^x[\text{H}^+]^y$ then calculate the value of $(x + y)$.
- A, B and C are isodiaphers while, C, D and E are isobars. Calculate the difference of protons between A and E ${}_{82}^{206}\text{A} \rightarrow \text{B} \rightarrow \text{C} \rightarrow \text{D} \rightarrow \text{E}$
Given : Isodiapheres and isobars are formed in successive α and β -emission respectively.
- In the given radioactive disintegration series
 ${}_{90}^{232}\text{Th} \rightarrow {}_{82}^{206}\text{Pb}$
Calculate value of $(n + 2)$
Where value of n is number of isobars formed in this series, suppose there is successive emission of β -particles.

9. In the given radioactive disintegration series



Calculate difference between number of α and number of β -particles in this series